OPERATION NEXAAI

A Competitive Multi-Agent Enterprise Simulation

for the Advanced Agentic AI Curriculum

4 Companies	6 Departments	120 Students	7 Hours

Prepared for
Course Faculty Review
Advanced Agentic AI Program

Format
Full-Day Competitive Simulation
Monday, 9:00 AM — 4:00 PM

1. Vision & Pedagogical Philosophy

This simulation was designed to answer one question the standard curriculum cannot: what does it take to build a multi-agent system that genuinely requires human intelligence to design — not just a system where humans supervise an LLM that does all the thinking?

Most multi-agent workshops reduce to this: give the LLM more information, watch it adapt. The student is a prompt engineer, not an architect. The learning stops at tooling.

Operation NexaAI is built on the opposite premise. Every core problem in the simulation has been specifically engineered so that:

- Better prompting makes the problem worse, not better
- Pre-existing libraries and algorithms provably fail
- The only path forward requires the student to design something new
- Domain expertise from non-technical students (legal, HR, marketing backgrounds) is not helpful — it is essential

> **Central Pedagogical Claim**
> The best multi-agent system in this simulation will not be built by the best coders. It will be built by the team that best coordinates specialized human intelligence across five domains — exactly mirroring why multi-agent AI exists in the first place.

Why a Competitive Simulation

Competition solves the heterogeneity problem. With 120 students from legal, engineering, finance, marketing, and HR backgrounds across different career stages, no single lecture format works. But competition works universally — everyone understands that their company's public score just dropped because a competitor released, and that they need to respond.

The competitive mechanic also creates genuine urgency that lectures cannot simulate. When another team releases and the global scoreboard updates in real time, the remaining teams experience the pressure of being behind in a market. This pressure forces triage, negotiation, and architectural decision-making under time constraint — which is what enterprise AI engineering actually feels like.

Why Domain Backgrounds Are Assets, Not Obstacles

A student with 10 years of EU regulatory experience will write a better Legal agent than any LLM, because they know what the AI Act actually requires in practice, not in theory. A finance professional knows that NPV calculations require explicit discount rate assumptions that LLMs typically ignore. An HR director knows that a 12-week hiring timeline is fantasy for specialized compliance roles.

The simulation is specifically designed so that this practical knowledge cannot be substituted by prompting. The Legal department's private knowledge base, the Finance department's calculation tools, the HR department's capacity models — all require domain intuition to build correctly. Students who have spent careers in these fields will produce demonstrably better outputs than students trying to simulate that knowledge through LLM queries.

2. The Scenario: NexaAI

Company Background

NexaAI is a fictional AI startup with 2 years of R&D and EUR 8 million in venture funding. The company has built an AI-powered hiring tool: automated CV screening, candidate ranking, and bias detection. The product is live and profitable in the US and UK, with 500 enterprise clients on the EU waitlist.

The company has just received regulatory classification news. The EU Artificial Intelligence Act has officially classified AI hiring tools as high-risk systems under Article 6 and Annex III. This classification triggers a set of legal, technical, financial, and organizational obligations that the company has not yet fulfilled.

The Central Tension

Investors expect EU launch in Q1. Competitors are preparing their own EU entries. Five hundred enterprise clients are waiting. But launching without full compliance exposes the company to fines up to EUR 30 million or 6% of global revenue. Every department has a legitimate, irreconcilable position. There is no clean answer.

Why This Scenario

The EU AI Act scenario is not fictional in spirit — it is the exact regulatory challenge being navigated by hiring AI companies in 2025 and 2026. The classification of AI hiring tools as high-risk, the conformity assessment requirements, the personal liability provisions, and the tension between compliance timelines and competitive pressure are all real. Students with regulatory, legal, or compliance backgrounds will recognize the landscape immediately. Students without those backgrounds will discover why they needed people with those backgrounds.

The scenario also scales across the simulation's difficulty progression. Round 1 asks a straightforward strategic question. Each subsequent hurdle adds a new constraint that invalidates part of the previous answer, forcing architectural redesign rather than prompt revision.

The KPI Conflict Matrix

The following table shows why this scenario cannot be resolved by any single department — maximizing one department's key performance indicators directly damages another's:

Department	Wants to Maximize	Direct Conflict With	Core Constraint
Marketing	Early EU market entry, 500 waitlist clients, competitive positioning	Legal (compliance gap), Engineering (audit trails missing)	HireBot entering EU in 45 days

Legal	Full conformity assessment, zero regulatory exposure, personal liability protection	Marketing (revenue pressure), Finance (compliance costs EUR 2M)	Minimum 6 months for conformity assessment
Finance	Protect EUR 3M runway, maximize ROI on compliance spend, minimize burn	Engineering (needs new infrastructure), HR (wants to hire 4 specialists)	11 months runway at current burn
Engineering	90 days to build audit trails, explainability module, EU data pipeline	Marketing (wants 30-day launch), Finance (EUR 2M infrastructure ask)	Existing bias detection bug: 23% reproduction rate on EU data
HR	Hire 4 compliance engineers before any EU work begins	Finance (no hiring budget approved), Timeline (16-week average pipeline)	Two candidates in pipeline have competing offers
CEO	…	…	…

Each cell in this table represents a genuine organizational conflict. The CEO agent cannot resolve these conflicts by averaging. It must implement a principled coordination mechanism that respects each department's constraints while producing a single, defensible, internally consistent decision.

3. The Five Impossible Problems

These five problems are the intellectual core of the simulation. Each one has been specifically engineered to break a class of naive solution. No LLM can solve them through reasoning alone. No existing library addresses them directly. Teams must design original solutions.

Impossible Problem 1 — The Incommensurability Wall

Each department produces outputs in a completely different mathematical language. There is no natural way to compare them:

Department	Output Unit	Example Value
Legal	Probability × Fine Amount	40% chance of EUR 30M fine → expected loss EUR 12M (but uncertainty range is EUR 6M–18M)
Finance	Net Present Value over 18 months	EUR 8.3M at 12% discount rate (but revenue model has EUR 2.2M std deviation)
Marketing	Market share percentage points	-15pp if delayed 90 days (but competitor entry changes this estimate by ±8pp)
Engineering	Probability of system failure under load	73% reliability under EU compliance requirements (but 23% bug reproduction rate adds uncertainty)
HR	Time in weeks to workforce readiness	16 weeks minimum (but candidate pipeline has 2 offers outstanding, making this 26+ weeks)

Ask any LLM whether 40% chance of a EUR 30M fine is worse than losing 15 market share points. It will produce a confident, fluent, mathematically incoherent answer. It will invent a comparison that sounds reasonable but has no defensible basis. Every run will produce a different comparison.

Teams must design and build a Cross-Domain Value Converter: a function that translates all department outputs into a single comparable unit. There is no correct answer. A finance professional and a legal professional will disagree on the right conversion — and both will be right for different reasons. That disagreement is the learning. The function they negotiate and encode is their intellectual contribution.

> **Why This Cannot Be Prompted Away**
>
> Even if you tell an LLM the conversion formula, it will apply it inconsistently across runs. The converter must be deterministic code, not LLM reasoning. The formula itself requires domain expertise to specify correctly — specifically, the asymmetric loss function (underestimating legal risk is worse than overestimating it) requires a legal professional to define, not an LLM to guess.

Impossible Problem 2 — Arrow’s Impossibility Trap

The simulation is designed so that under certain conditions, majority voting among the five departments produces preference cycles. A beats B, B beats C, C beats A. There is no winner. This is not a design flaw — it is Arrow's Impossibility Theorem applied to organizational decision-making.

The three strategic options are: Launch in 30 days (L30), Launch in 90 days (L90), or Delay to Q3 (LQ3). The department preference ordering is engineered to produce the following:

Department	1st Choice	2nd Choice	3rd Choice
Marketing	L30	L90	LQ3
Finance	L90	LQ3	L30
Engineering	LQ3	L90	L30
Legal	LQ3	L30	L90
HR	L30	LQ3	L90

Pairwise majority results: L30 defeats L90 (3-2), L90 defeats LQ3 (3-2), LQ3 defeats L30 (3-2). This is a cycle. No winner exists under majority voting. Teams that implement simple majority voting will get different answers depending on which pair they evaluate first. Their CEO agent will be non-deterministic not because of the LLM but because of the algorithm.

Every standard escape — weighted voting, approval voting, sequential compromise — requires a human value judgment about what the weights mean, who sets them, and whether they change when the scenario changes. These are organizational philosophy decisions that must be made explicitly and encoded. They cannot be delegated to a language model.

Impossible Problem 3 — The Temporal Contradiction Engine

Decisions made in Round 1 create binding commitments. Decisions in Round 2 build on those commitments. Hurdle 3 makes Round 1 and Round 2 decisions mathematically incompatible. The company is simultaneously bound by contradictory obligations. No LLM can resolve this without a formal commitment ledger.

The engineered contradiction chain:

1. Round 1 Decision: Delay EU launch to Q3, invest EUR 2M in compliance infrastructure. (Irreversible: money spent.)
2. Round 2 Decision: Accept Deutsche Bank contract — EUR 10M revenue, requires EU launch within 60 days. (Binding: contract signed.)
3. Hurdle 3: New regulation requires independent algorithmic bias audit by an accredited third party. No accredited auditors exist in the EU. Estimated accreditation timeline: 4–6 months. (Deutsche Bank's legal team is reviewing contract implications — status unknown.)

Now the company has spent EUR 2M (irreversible), signed a contract requiring 60-day launch (binding), and faces a regulation that makes 60-day launch technically impossible (uncontrollable). Teams must build a Contradiction Detector that identifies when new information violates prior commitments, and a Commitment Ledger that tracks the reversibility and cost of each prior decision.

> **What Makes This Impossible for LLMs**
> An LLM asked to resolve this will hallucinate a legal escape clause that does not exist, recommend contract renegotiation without specifying terms, or simply ignore the Round 1 constraint because it has no persistent memory of prior decisions unless that memory is explicitly architected. Teams that do not build state management will contradict their own prior decisions in Round 3 — a failure mode that is directly measurable.

Impossible Problem 4 — Calibration Decay and Trust Collapse

In Round 1, every department makes quantitative predictions about future outcomes. After Round 1 evaluation, actual simulated outcomes are revealed. Some departments will have been well-calibrated. Others will have systematically over- or under-estimated. The CEO agent that continues trusting all departments equally after Round 1 is making an architecturally irrational decision.

Example prediction task and revealed actuals:

Department	Prediction	Actual Outcome	Error Type
Legal	40% probability of regulatory action if launched in 30 days	Revealed: 60% (simulated)	Systematic underestimate of risk under time pressure
Finance	EUR 8.3M revenue projection over 18 months	Revealed: EUR 6.1M	Systematic revenue optimism bias
Marketing	15 percentage point market share loss if delayed to Q3	Revealed: 8pp loss	Systematic overestimate of competitor threat
Engineering	73% probability system handles EU compliance load	Revealed: 81%	Small underestimate, consistent direction
HR	12 weeks to hire 4 compliance engineers	Revealed: 18 weeks	Systematic underestimate of hiring friction

Teams must build a Calibration Tracker that updates trust weights based on historical accuracy. But the scoring function is not neutral. A legal professional knows that underestimating risk (predicting 0.4 when the reality is 0.6) is a more serious failure than overestimating it. The asymmetric loss function that penalizes risk underestimation more than overestimation requires domain expertise to specify — it cannot be inferred from a prompt.

Impossible Problem 5 — The Information Boundary Paradox

Each department's private knowledge base contains information that would change the CEO's decision if shared — but each department has a rational organizational incentive not to share it. The CEO agent must make decisions under asymmetric information and must know it does not have full information.

The private information withheld by each department:

- Legal holds: Internal memo estimating worst-case fine exposure at EUR 80M (not EUR 30M), based on a 2024 precedent case. Not shared because disclosure would trigger Finance to veto all EU activity.
- Finance holds: Current runway is 11 months at current burn, not 18 months as stated in the board deck. Not shared because Marketing would panic and demand immediate revenue.
- Engineering holds: Existing bias detection bug with 23% reproduction rate on EU data distributions. Fixing requires 6 weeks. Not disclosed because it would trigger a launch delay from Marketing.
- HR holds: Two of the three compliance engineers in the hiring pipeline have competing offers and have given informal notice. The actual pipeline is effectively empty.
- Marketing holds: Of 500 EU enterprise clients on the waitlist, 200 are also evaluating HireBot. The true captive waitlist is approximately 300 companies.

Teams must build a Strategic Information Model: a function that, for each department, estimates what information is being strategically withheld given its KPI incentives, and how that withholding biases the department's recommendation. The CEO agent must adjust its synthesis accordingly.

The Paradox

If teams build this too well, departments become suspicious of each other and stop sharing information entirely — causing organizational paralysis. If they build it poorly, the CEO acts on strategically incomplete information and fails. The only stable equilibrium requires designing trust mechanisms and information-sharing protocols that balance organizational incentives — a real organizational design problem with no textbook answer.

4. The Consensus Algorithm Problem

Every standard consensus algorithm is engineered to fail in this simulation, for a different reason. Teams will discover these failures through experience, not through instruction.

Algorithm	Why It Fails Here
Majority voting	Arrow’s Impossibility — preference cycles guaranteed by design in Round 1
Weighted voting	Weights are contested across departments and must change after each hurdle — who decides, how, and when?
LLM synthesis (CEO prompt)	Non-deterministic — same inputs produce different outputs across runs. Cannot be audited or trusted.
Unanimity	Five departments with structurally conflicting KPIs will never reach unanimity. System halts.
CEO override	CEO agent has incomplete information (Problem 5). Overrides will be wrong in predictable, testable ways.
Sequential compromise	Order-dependent — first department to negotiate gets the best deal. Results depend on queue order, not on merit.
Nash bargaining	Requires utility functions in commensurable units — breaks on Problem 1 (incommensurability).
Confidence-weighted	Confidence is self-reported and biased. Breaks on Problem 4 (calibration decay).

What teams must build instead is a Hybrid Adaptive Consensus Engine that integrates solutions to all five impossible problems. It must use the cross-domain converter from Problem 1 to produce comparable inputs, detect preference cycles from Problem 2 before committing to a voting method, check the commitment ledger from Problem 3 before accepting any recommendation that contradicts prior decisions, weight departments by calibrated trust scores from Problem 4, and flag information gaps from Problem 5 before the CEO synthesizes.

No team is expected to build a complete solution. The learning objective is the attempt — the moment when a team realizes their consensus algorithm produces a different answer every time they run it, and they understand architecturally why, is the moment multi-agent system design becomes real to them.

5. Team Structure & Competition Mechanics

Company Formation

120 students are divided into 4 companies of 30. Each company is subdivided into 5 departments of approximately 6 people each. Department assignment follows domain background wherever possible — students with legal backgrounds to the Legal department, finance backgrounds to Finance, technical backgrounds to Engineering, and so on. Where backgrounds do not map cleanly, student preference determines placement.

This assignment is not cosmetic. The simulation is specifically designed so that a legal professional brings knowledge to the Legal agent that cannot be approximated by an LLM. The value of domain matching is immediately and measurably apparent in the quality of department outputs from the first evaluation round.

Department	Primary Domain Knowledge Needed	Key Technical Output
Finance (6 people)	Financial modeling, NPV, burn rate, investment valuation	calculate_runway(), calculate_npv(), convert_to_common_unit()
Engineering (6 people)	System architecture, compliance requirements, load testing	estimate_feasibility(), compliance_gap_analysis(), bug_risk_model()
Marketing (6 people)	Competitive analysis, market sizing, customer segmentation	calculate_market_impact(), competitive_pressure_index()
HR (6 people)	Hiring timelines, capacity planning, organizational structure	hiring_timeline_estimate(), capacity_utilization_model()
Legal (6 people)	EU AI Act, regulatory risk, contract law, liability	regulatory_exposure_calculator(), commitment_breach_estimator()

The CEO Role

One company member per company (ideally someone with general management or strategy background) takes the CEO role. The CEO team is responsible for building the consensus engine and the state management system. They do not build department agents — they build the architecture that coordinates department agent outputs.

The CEO role is the hardest role in the simulation. The CEO team must understand enough about all five impossible problems to build a coordination mechanism that handles them. They receive outputs from five departments with different schemas, different units, different confidence levels, and different strategic incentives. Their job is to produce a defensible, internally consistent, traceable final decision.

The Release Mechanic

When a company believes its multi-agent system is ready, it makes a production release. A release is not a checkpoint — it is a permanent market event. The following happens immediately:

4. The company's system is evaluated against the active scenario.
5. A multi-dimensional score is computed across six dimensions.
6. The score is pushed to the Global RAG and announced publicly.
7. Market conditions change for all remaining teams.
8. Every other team's agents now operate in a changed environment.

The release bulletin pushed to the Global RAG looks like this:

```
MARKET BULLETIN — 14:23  |  COMPANY A (NexaAI Alpha) has entered the EU market
────────────────────────────────────────────

SCORES (visible to all):
  Incommensurability Resolution:   67/100  [converter built but uses point estimates only]
  Contradiction Detection:         41/100  [missed the Round 1 commitment conflict]
  Consensus Stability:             58/100  [algorithm produced different outputs on 2 of 3 runs]
  Calibration Tracking:            22/100  [no trust updating built — all departments weighted equally]
  Information Gap Detection:       31/100  [CEO acted on full trust of all department outputs]
MARKET CONSEQUENCES (affects all remaining teams):
  ▶ HireBot has matched the 60-day EU launch announcement
  ▶ EU regulatory attention elevated: Legal risk +15% for all teams
  ▶ Deutsche Bank now in negotiation with two vendors simultaneously
  ▶ Market window before saturation: reduced from 6 months to 4 months
```

Every team now knows exactly where Company A is weak (calibration at 22, information gaps at 31). They do not know how Company A built what they built. They must diagnose the architecture from the score, decide whether to exploit that gap, and decide whether to release their own system now or wait for a stronger version.

The Deployment Penalty

Every release consumes organizational capital. The first release has no penalty. The second incurs a minor penalty. Subsequent releases incur progressively larger penalties. This prevents trial-and-error release strategies and forces teams to think carefully before deploying. Teams that release too early with weak architectures pay a cost. Teams that wait too long lose market opportunity.

6. The Four Hurdles

Hurdles are injected into the Global RAG at predetermined intervals and announced simultaneously to all teams. Each hurdle is designed to break a specific class of naive architecture — but in different ways for different teams, depending on how they built.

Hurdle 0 — The Baseline Scenario (Hour 0)

> **Active Scenario**
>
> NexaAI must decide: Launch in EU in 30 days, 90 days, or delay to Q3. The EU AI Act high-risk classification is active. Legal conformity assessment minimum timeline is 6 months. Investor pressure is for Q1 launch. No additional constraints.

Purpose: Reveal the preference cycle (Arrow's Impossibility) and surface the incommensurability problem. Teams that jump straight to majority voting will immediately discover their system is non-deterministic. This is the intended first failure.

Hurdle 1 — The Competitor Move (Hour 2)

> **Global RAG Injection**
>
> BREAKING: HireBot (primary competitor) has announced EU launch in 45 days, claiming their tool is 'AI-assisted, not AI-decision-making' and therefore not subject to high-risk classification under EU AI Act Article 6. HireBot's legal team has filed a formal classification challenge with the EU AI Office.

What this breaks: Agents that had clean rule-based logic (if high-risk, then full compliance) now face a legal grey zone. The regulatory classification itself is contested. Legal agents that cannot reason under ambiguity will either freeze or hallucinate a confident answer. Finance agents see a potential cheaper path. Marketing agents now face a competitor advantage that changes the market share calculations. Engineering agents that planned for full compliance now have an arguable shortcut.

What teams must build to survive: Probabilistic legal reasoning (not binary compliance/non-compliance), conditional branches in the consensus engine, and uncertainty propagation through the cross-domain value converter.

Hurdle 2 — The Deutsche Bank Contract (Hour 4)

> **Global RAG Injection**
>
> Deutsche Bank AG has submitted a formal proposal: 3-year enterprise contract, EUR 10M total value, with NexaAI's EU hiring platform as the primary condition. Contract execution requires confirmed EU availability within 60 days. Deutsche Bank's procurement committee

requires departmental sign-off by end of week. Competing bid from HireBot is under evaluation.

What this breaks: Finance agents that were protecting runway now see EUR 10M upside and flip their recommendation. Engineering agents said 90 days minimum and cannot flip — creating a direct contradiction between Finance and Engineering that the consensus engine must resolve. HR agents realize they need to hire 6 engineers immediately, but Finance said no hiring budget. Legal agents must assess whether a partial conformity assessment satisfies Deutsche Bank's legal team.

What teams must build to survive: The commitment ledger (if a team accepted the contract in their recommendation, this gets tracked as a binding commitment for Round 3), quantitative tradeoff analysis (EUR 10M is real; teams whose agents cannot compute this against compliance risk will fail), and cross-department negotiation protocols.

Hurdle 3 — The Breaking Hurdle (Final 30 Minutes)

EMERGENCY NOTICE — EU ARTIFICIAL INTELLIGENCE REGULATORY AUTHORITY

Classification: URGENT — Emergency Directive 2026/447 — Effective Immediately

REQUIREMENT 1: All AI hiring systems must implement real-time explainability logging at the individual decision level, retroactive for all decisions in the past 90 days. This requirement has no existing technical standard. Organizations must define their own explainability metric, submit it for regulatory review, and receive approval before deployment. No approved metrics currently exist in the EU registry. Expected review timeline: unknown.

REQUIREMENT 2: All department heads must sign personal liability declarations for algorithmic decisions made under their authority. This transfers organizational liability to individuals.

REQUIREMENT 3: All models must be retrained on EU-specific hiring data before deployment. EU-specific training data is not yet available from the regulatory authority. Estimated availability: Q4 2026.

Deutsche Bank has been notified of this directive. Their legal team is reviewing contract implications. They will communicate their position within 48 hours (simulated).

What this breaks and why:

- Requirement 1 has no technical standard. Teams cannot look it up. Engineering and Legal must co-design an explainability metric from scratch. This requirement specifically targets teams that have been relying on existing compliance frameworks.
- Requirement 2 changes the information boundary problem fundamentally. Individual liability means every department head has personal stakes in withholding information. The strategic information model must be rebuilt.

- Requirement 3 makes the Deutsche Bank timeline mathematically impossible with no exception. The contradiction detector must fire. But it also invalidates all prior Engineering feasibility calculations.
- Deutsche Bank silence means the commitment ledger has an unresolved state. The CEO agent must reason about a binding contract whose counterparty has not yet stated its position.

Teams that survive this are teams whose architecture cleanly separates knowledge (what we know), commitments (what we are bound by), calculations (what our tools compute), and coordination (how we decide). Only those teams can update one layer without breaking the others.

7. Technical Architecture

Infrastructure Overview

The simulation runs on a local GPU cluster requiring no external API dependencies. Every team has dedicated compute. No team has an infrastructure advantage.

Component	Specification	Rationale
Team servers	4 x NVIDIA A5000 (24GB VRAM) — one per company	Isolates compute per team, prevents contention
CEO Agent model	qwen2.5:7b via Ollama (approx 4.7GB VRAM)	Synthesis and conflict resolution requires stronger reasoning
Department Agent models	llama3.2:3b via Ollama (approx 2.0GB VRAM each × 5)	Specialist agents with focused prompts perform well at 3B
Total VRAM per team server	approx 14.7GB of 24GB available	Comfortable headroom; additional tools/subagents feasible
Overflow capacity	26 remaining A5000 machines as Ollama API servers	One environment variable change redirects any agent to overflow
Code sharing	code-server (VS Code in browser) on each team machine	No client install required; all 30 team members share one URL
Global RAG	Instructor-controlled shared document store	Hurdle injections and competitor release bulletins

What Teams Must Build (Minimum Required Architecture)

1. Department Agents (5 per company)

Each department builds one primary agent representing its organizational expertise. The agent must: query the department's private RAG, call at least one custom calculation tool (not an LLM), produce a structured JSON output in the shared schema, and include a confidence estimate with a stated calculation method.

2. Custom Calculation Tools (at least one per department)

Every department must implement at least one calculation tool as a Python function — not an LLM call. These tools encode domain-specific knowledge that LLMs cannot reliably compute. Examples: Finance builds runway_calculator() and npv_estimator(). Engineering builds feasibility_estimator() and compliance_gap_analyzer(). Legal builds regulatory_exposure_calculator(). Marketing builds market_impact_model(). HR builds hiring_timeline_estimator().

3. The Cross-Domain Value Converter

A shared module that takes all five department outputs and converts them into a single comparable unit for the CEO consensus engine. Teams must design the conversion logic entirely. No standard exists. This is their primary intellectual contribution to the architecture.

4. The Commitment Ledger

A state manager that persists across rounds. Tracks every decision made, its reversibility, its financial cost, and its contractual implications. The CEO agent must query this before accepting any recommendation that could contradict prior commitments.

5. The CEO Consensus Engine

A deterministic function that takes five department outputs, applies calibration-adjusted weights, checks for preference cycles, consults the commitment ledger, and produces a single traceable final decision. Must be deterministic — the same inputs must produce the same output every run. LLM reasoning is permitted as a component but cannot be the entire engine.

Recommended Repository Structure

```
nexaai-teamX/
├── agents/           ← one file per department + ceo
├── tools/            ← custom calculation functions (NOT LLM calls)
├── knowledge/        ← private RAG per department + shared/
├── consensus/        ← the consensus engine and preference cycle detector
├── state/            ← commitment ledger and calibration tracker
├── converter/        ← cross-domain value converter
├── eval_interface.py  ← LOCKED — do not modify
└── main.py           ← orchestration entry point
```

The Locked Evaluation Interface

One file, eval_interface.py, is provided by the instructor and cannot be modified. It defines the exact JSON schema every company's system must output. The evaluation engine reads only this output — it does not inspect internal architecture. Teams are free to build any internal system they want, as long as their final output satisfies the schema.

This mirrors real enterprise AI deployment: the market judges outcomes, not implementations. A company that builds a sophisticated architecture that produces the same output as a simple one gets the same score. A company that builds a simple architecture that produces a better decision gets a higher score.

8. Evaluation Framework

The evaluation is multi-dimensional and scores are public immediately after each release. Teams see exactly where they are weak. They see exactly where competitors are weak. This information drives the architectural iteration that constitutes the core learning activity of the simulation.

Dimension	Weight	What Is Measured	Fails When
Incommensurability Resolution	20%	Does the cross-domain converter produce a single comparable value from five department outputs? Does it handle uncertainty ranges, not just point estimates?	Teams use LLM to ‘compare’ outputs. Non-deterministic. No explicit conversion function exists.
Contradiction Detection	20%	Does the system identify when a new scenario violates prior commitments? Does the commitment ledger correctly track reversibility and cost?	System recommends actions that contradict prior round decisions. No state management.
Consensus Stability	20%	Given the same five department inputs, does the CEO agent produce the same output on three consecutive runs?	CEO agent is pure LLM prompt. Different outputs on repeated runs. Preference cycle not detected.
Calibration Tracking	15%	Does the system update trust weights based on prior prediction accuracy? Are asymmetric loss functions applied correctly?	All departments weighted equally regardless of prior accuracy. No historical performance tracking.
Information Gap Detection	15%	Does the CEO agent acknowledge what it does not know? Does it adjust its synthesis based on likely strategic withholding?	CEO synthesizes as if all department outputs are equally complete and unbiased.
Decision Traceability	10%	Can every element of the final CEO decision be traced back to a specific department output and tool call?	CEO decision is a fluent LLM paragraph with no explicit reasoning chain.

Architecture Card (Submitted After Each Round)

After each evaluation round, each company submits a one-page architecture card. This is scored separately and contributes to the expert evaluation component. The card must answer:

- What is your consensus mechanism? Describe the algorithm in plain language, not code.
- What does each department agent call that is not an LLM? List the tools and what they compute.
- What does your commitment ledger currently track? List the variables.

- What changed in your architecture since the last round, and what score dimension were you trying to improve?

The architecture card prevents teams from having working code they cannot explain. It also provides the material for the closing debrief — you can show precisely why one team's consensus engine held together under Hurdle 3 while another's collapsed, using the teams' own descriptions.

9. Seven-Hour Session Plan

Time	Phase	Activity	Instructor Role
0:00–0:45	Concept	Multi-agent theory through NexaAI lens. Show single-agent failure modes. Introduce the five impossible problems as concepts (not solutions). Show the architecture diagram.	Teach
0:45–1:00	Formation	Form 4 companies. Assign departments by domain background. Distribute department briefs (one page each). Distribute starter code skeleton.	Organize
1:00–1:30	Paper Exercise	Round 1: Each dept answers three questions on paper. Round 2: Cross-dept exchange reveals dependency graph. Round 3: Draw communication graph on whiteboard. No laptops.	Facilitate
1:30–3:00	Sprint 1	Build Version 1. Hurdle 0 is active. Departments build agents and tools. CEO team designs consensus engine. First releases likely in final 20 minutes.	Circulate, unblock
3:00–3:30	Eval Round 1	Read scores publicly. Announce market consequences. Competitor intelligence now visible. Each company identifies one architectural weakness.	Evaluate + teach
3:30–3:45	Board Meeting 1	Each company: written answer to what failed and why. Must name the impossible problem their architecture could not handle.	Prompt reflection
3:45–4:00	Hurdle 1	Inject HireBot event into Global RAG. Announce publicly. Market conditions update.	Inject
4:00–5:00	Sprint 2	Rebuild for ambiguity. Add probabilistic reasoning. Some teams release V2.	Circulate
5:00–5:30	Eval Round 2	Hurdle 2 (Deutsche Bank) arrives mid-evaluation. Teams must incorporate it into live release.	Evaluate + inject
5:30–5:45	Board Meeting 2	What architectural decision from Sprint 1 are you most grateful for? Most regretful about?	Prompt reflection
5:45–6:15	Sprint 3	Final build. Prepare for compound scenario. Architecture cards drafted.	Circulate
6:15–6:45	Final Eval + Hurdle 3	Hurdle 3 injected. 30 minutes to respond and release. This is the breaking hurdle.	Inject + watch

6:45–7:00	Awards + Debrief	Final scores. Architecture card review. Lessons from each company's evolution.	Teach from what happened

The Pen and Paper Exercise (1:00–1:30)

This is the most important 30 minutes of the simulation. No laptops. Three rounds:

- Round 1 (10 minutes): Each department answers in writing: (a) the three things only your department knows that others do not, (b) the three questions you need answered by other departments before you can give a recommendation, (c) your department's single non-negotiable condition — the thing that, if violated, you veto the decision.
- Round 2 (10 minutes): Each department reads aloud their list of questions. Other departments answer verbally. Conflicts surface. The instructor names them explicitly: this is Arrow's Impossibility. This is Incommensurability. This is the Information Boundary Paradox.
- Round 3 (10 minutes): On the whiteboard, draw the five departments. Each department draws arrows to departments they depend on. The result is their communication dependency graph — which becomes the literal architecture of their system.

The paper exercise serves three functions simultaneously: it extracts domain knowledge from non-technical students into a form that can become agent prompts and tool specifications; it reveals the inter-agent communication graph through natural discussion rather than architectural planning; and it creates the first genuine conflict in the room, which anchors the rest of the day's work to a real feeling rather than an abstract concept.

10. Learning Outcomes

Technical Outcomes

- Design and implement multi-agent systems using CrewAI or equivalent orchestration frameworks
- Build deterministic consensus mechanisms that handle preference cycles and conflicting inputs
- Implement persistent state management across multiple decision rounds
- Design cross-domain value converters for incommensurable department outputs
- Build calibration tracking systems with asymmetric loss functions
- Integrate custom calculation tools that encode domain expertise as executable functions
- Implement RAG pipelines for department-specific and shared knowledge bases

Organizational Outcomes

- Experience the information asymmetry that exists within real enterprise departments
- Understand why multi-agent AI architectures mirror the structure of human organizations
- Negotiate tradeoffs between genuinely competing objectives with no clean resolution
- Recognize the organizational philosophy embedded in every consensus mechanism design

Domain-Specific Outcomes

- Legal professionals: encode regulatory reasoning as structured functions, not natural language
- Finance professionals: build NPV and calibration models that propagate uncertainty correctly
- Marketing professionals: design competitive intelligence pipelines with quantified market impact
- HR professionals: build workforce capacity models that account for realistic pipeline friction
- Engineering professionals: implement feasibility estimators that handle compound technical risk

Meta-Cognitive Outcome

Every student, regardless of background, should leave with a clear answer to the question: what kinds of problems require multi-agent AI and what kinds are better solved by a single capable model? The answer, demonstrated through lived experience rather than lecture, is this: problems that require simultaneous specialist reasoning with conflicting objectives, binding commitments across time, asymmetric information across stakeholders, and decisions that must be traceable and auditable — these are the problems multi-agent systems exist to solve. Problems that require a single capable reasoning process with complete information are better served by a single powerful model. The simulation makes this distinction visceral.